Learning Sparse Pure States with Single-Qubit Pauli Measurements

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Abstract

We learn an unknown n-qubit pure state with at most k nonzero computational-basis amplitudes using only single-qubit Pauli measurements. The learner fixes all bases on a copy before reading any outcome. An initial computational-basis sample estimates magnitudes. To recover phases, the learner compares pairs of amplitudes along a shallow tree. A comparison receives more copies when its signal is weak or when the labels that depend on it carry large total probability. The tree keeps the total comparison weight at each level linear in k. This gives infidelity at most ε with probability at least $1-\delta$ using $\widetilde{O}(k^2/\varepsilon)$ copies and $\widetilde{O}(nk^2/\varepsilon)$ time.

1 Introduction

A source supplies independent copies of an unknown n-qubit pure state. The learner knows that at most k computational-basis strings have nonzero amplitude. Measuring each qubit in Z gives a string x with probability $p_x = |a_x|^2$. Repeating the measurement reveals which strings occur in the sample and estimates their probabilities. Even infinitely many such measurements would leave every relative phase undetermined, since changing a phase leaves every p_x unchanged.

How many copies suffice to recover the state using only single-qubit X, Y, Z measurements? Sen achieves $\widetilde{O}(k/\varepsilon)$ copies and $\widetilde{O}(nk/\varepsilon)$ time using CNOT circuits followed by Pauli measurements [Sen26]. Here the learner uses no CNOTs or other entangling gates. Earlier copies may guide later measurements, but the learner fixes all bases on a copy before reading any of its outcomes.

We show that these measurements suffice for every support, even when some labels are too rare to appear. Up to logarithms, our upper bounds have one additional factor of k compared with Sen's benchmark. The output lists the labels and their estimated amplitudes, up to a common phase.

Theorem 1. Learning a sparse pure state. Given copies of $|\psi\rangle = \sum_{x\in S} a_x|x\rangle$, the qubit count n, a bound $1 \le |S| \le k \le 2^n$, and $0 < \varepsilon, \delta < 1$, there is a learner in the single-qubit Pauli model above with the following guarantee. The support, probabilities and phases are unknown. With probability at least $1-\delta$, its normalized output $|\widehat{\psi}\rangle$ has infidelity

$$1 - |\langle \psi \mid \widehat{\psi}\rangle|^2 \le \varepsilon.$$

It outputs at most k amplitudes, uses $\widetilde{O}(k^2/\varepsilon)$ copies and takes $\widetilde{O}(nk^2/\varepsilon)$ total time. Section 5 gives the logarithmic factors.

From magnitudes to phases. Suppose we can estimate the phase of one amplitude relative to another. Exact comparisons with a single reference would determine every relative phase. With finite data, a rare reference gives a weak signal each time it is used. Moving the reference closer to each group of labels can improve the comparisons, but an error made on the way into a group affects the whole group. The construction below controls both costs. We first determine what a single comparison can measure, then choose how to connect the labels and how many copies to spend on each connection. The first batch of Z measurements supplies all the information needed to make these choices on an unknown state.

Related work. For computational-basis sparse states, Gulbahar proposed a phase-estimation method that assumes efficient circuits for large powers of a specially chosen unitary [Gul21]. Sen asks whether his near-linear runtime can be attained with single-qubit Pauli measurements alone [Sen26, Sec. 1.4]. At the dense endpoint $k = 2^n$, Grewal et al. achieve $\widetilde{O}(2^n/\varepsilon)$ copies and time using nonadaptive Pauli measurements [GGH$^+$26]. For sparse supports, Li, Wu, and Zhang connect amplitudes along a spanning tree and count $1 + 2(k-1)$ measurement settings, with an entanglement-free implementation [LWZ25]. A setting can be repeated on many copies. Our bound counts the copies needed to recover the phases at a prescribed infidelity. Lee, Jiang, and Sharma estimate selected density-matrix entries with single-qubit measurements [LJS26]. Here the variance of each selected entry determines its share of the copy budget.

Related uses of single-copy data have different outputs. Grier, Pashayan, and Schaeffer construct classical shadows of pure states to predict observable expectations. Their independent-copy protocol can use a global random Clifford measurement on each copy [GPS24, Sec. 5]. In certification, the verifier receives a description of the pure state it should expect. Gupta, He, and O'Donnell give a test for every pure target using adaptive single-qubit measurements [GHO25]. Coladangelo, Li, and Slote give a robust test using nonadaptive single-qubit Pauli measurements for all but an exponentially small Haar fraction of pure targets [CLS26, Cor. 1]. Our learner receives only a sparsity bound and must construct the amplitude list itself.

2 What can one Pauli comparison tell us?

Temporarily give the learner the support S and magnitudes $\sqrt{p_x}$. Write $a_x = \sqrt{p_x}\, e^{i\theta_x}$. For two distinct labels $x, y \in S$, the complex number

$$\mu_{xy} = \overline{a_x} a_y = \sqrt{p_x p_y}\, e^{i(\theta_y - \theta_x)}$$

contains exactly the phase change needed to get from x to y. Its direction $z_{xy} = \mu_{xy}/|\mu_{xy}|$ is a unit complex number. Multiplying the phase at x by z_{xy} gives the phase at y. How can a product measurement estimate this direction when the state may contain many other labels?

Follow one copy. Compare the bits of x and y. Where they agree, choose Z. At each position in $J = \{j : x_j \neq y_j\}$, independently choose X or Y with equal probability. Fix all these bases, measure the copy, and give it one complex score S_{xy}. If any Z result differs from the common bit, record zero. Otherwise multiply the following factors over the differing positions. A sign $\sigma_j \in \{+1, -1\}$ contributes σ_j for an X measurement and $i(y_j - x_j)\sigma_j$ for a Y measurement. For instance, if only position $t \in J$ uses Y and the check passes, that record contributes $i(y_t - x_t)\prod_{j \in J} \sigma_j$. The learner averages all scores, including zeros.

Let $\mathcal{B}_{xy}$ contain all bit strings agreeing with x on the positions outside J. The probability of passing the Z check is $P_{xy} = \sum_{b \in \mathcal{B}_{xy}} p_b$, with $p_b = 0$ outside S. Other support strings may pass.

They enter the second moment through P_{xy}, while averaging the signed X, Y outcomes selects just the desired pair in the mean. These two facts determine how accurately we can estimate its phase.

Lemma 1. Measuring one relative phase. For the product measurement above and distinct $x, y \in S$, $\mathbb{E}S_{xy} = \mu_{xy}$ and $\mathbb{E}|S_{xy}|^2 = P_{xy}$. From $s \geq 1$ independent copies form their mean $\bar{S}$ and set $\widehat{z}_{xy} = \mathrm{ph}(\bar{S})$, where $\mathrm{ph}(v) = v/|v|$ for nonzero v and $\mathrm{ph}(0) = 1$. Then

$$\mathbb{E}|\widehat{z}_{xy} - z_{xy}|^2 \leq \frac{4P_{xy}}{s\, p_x p_y}.$$

Proof. For a measurement with outcomes labeled by scores, the expected score is the expectation of the sum of its outcome projectors multiplied by their scores. On an agreeing bit, only the matching outcome contributes, so the sum is $|x_j\rangle\langle x_j|$. On a differing bit, weighting the two outcome projectors by their signs gives the measured Pauli. Averaging also over the fair basis choice gives

$$\tfrac{1}{2}\big(X + i(y_j - x_j)Y\big) = |x_j\rangle\langle y_j|.$$

Indeed $(X + iY)/2 = |0\rangle\langle 1|$ and $(X - iY)/2 = |1\rangle\langle 0|$. Multiplying scores takes tensor products of the local operators. The independent basis choices let us take their averages one coordinate at a time, without assuming independent outcomes on an entangled state. The resulting operator is $|x\rangle\langle y|$, so its expectation in the full state is $\overline{a_x} a_y$. All other pairs of amplitudes have zero coefficient in this expectation.

On a pass the score has magnitude one. Summing over all outcomes on the differing bits leaves the identity there, so passing depends only on the agreeing bits. Thus $\mathbb{E}|S_{xy}|^2 = P_{xy}$. Independence across copies gives

$$\mathbb{E}|\bar{S} - \mu_{xy}|^2 = \frac{P_{xy} - |\mu_{xy}|^2}{s} \leq \frac{P_{xy}}{s}.$$

To pass from a complex number to its direction, write $v = |v|\,\mathrm{ph}(v)$. The triangle inequality and $\big||v| - |\mu|\big| \leq |v - \mu|$ give

$$|\mu|\,|\,\mathrm{ph}(v) - \mathrm{ph}(\mu)| \leq \big||\mu| - |v|\big| + |v - \mu| \leq 2|v - \mu|.$$

Take $v = \bar{S}$, square, and use $|\mu_{xy}|^2 = p_x p_y$. □

Which comparisons recover all the phases? The cancellation on two consecutive comparisons follows from $z_{xy} z_{yv} = e^{i(\theta_y - \theta_x)} e^{i(\theta_v - \theta_y)} = z_{xv}$. Thus a path from a reference label to every other label suffices. Pereira, Zambrano, and Delgado likewise assemble larger amplitude blocks by estimating the relative phase between two previously reconstructed blocks from separable-basis measurements [PZD22]. In our setting the choice of paths matters with noisy estimates. Long paths accumulate several errors, while even a single comparison can be expensive if P_{xy} is large relative to $p_x p_y$. The next construction limits how many comparisons a path uses and how much they cost in total.

3 A shallow tree of affordable comparisons

Let U be a nonempty set of at most k available labels, with probabilities $q_x > 0$ summing to one. For now $U = S$ and $q = p$. Later the same construction will take observed labels and frequencies as input. For a group $C \subseteq U$, write $M_C = \sum_{x \in C} q_x$. Give each group a reference label. To learn a singleton's phase, start at the root reference and multiply the estimated ratios along the path to that singleton. The number of factors matters because an error in any one factor changes the product.

3.1 How deep must the tree be?

If a bit divides a group in half, both resulting groups are immediately smaller. On an arbitrary support even the best bit may put very few labels on one side. We can still make a shallow tree by letting several successive bit tests produce children of the *same* parent.

Lemma 2. A shallow partition by bit tests. For any nonempty set U of at most k distinct labels, there is a tree rooted at U with singleton leaves and at most $L = 1 + 2\lceil \log_2 k \rceil$ internal levels. Each group is the set of labels in U satisfying fixed bit tests. For a group C, let

$$t(C) = \max_j \min\{\, |\{x \in C : x_j = 0\}|,\ |\{x \in C : x_j = 1\}|\,\}.$$

Every child of C has at least $t(C)/2$ labels.

Proof. The smaller side of the best single-bit split of C contains $t(C)$ labels. Fix an internal parent C of size m, set $t = t(C)$, and start with all its labels in a list $R = C$. Split R on its best bit precisely when

$$|R| > m/2, \qquad t(R) \geq t/2.$$

Save the smaller side as a child of C and continue with the larger side as R. When a test fails, save the remaining R as the final child. Resolve ties by a fixed rule. The first split occurs because two distinct labels differ somewhere, so $t \geq 1$. Each split has at least $t/2$ labels on either side, including the side kept after the last split. Membership in a child is determined by the bit tests leading to it and those inherited from its parent.

Along any descent from C to a child D, either $|D| \leq m/2$, or D is the final remainder and the stopping rule gives $t(D) < t/2$. Restriction to a subset can increase neither its size nor either bit count. Hence $|D|t(D) \leq mt/2$. At the root both factors are at most k. After ℓ descents the product is at most $k^2/2^\ell$. At an internal node it is at least one, so $\ell \leq 2\log_2 k$. Counting the root level proves the claim. □

3.2 How many copies should one comparison receive?

For a parent C and child D, we allow two comparisons from their references, $h_C \to g \to h_D$, with $g \in D$. This lets us enter D through a label that is easy to compare with h_C. A step with identical endpoints has known ratio 1 and is omitted. Otherwise count each use as its own comparison $e = (x, y)$ on the link into D.

Every leaf below D uses that comparison. Its squared phase error is therefore weighted by M_D, the probability mass that inherits it. Write $Q_{xy} = \sum_{b \in U \cap \mathcal{B}_{xy}} q_b$ for the mass passing the agreeing-bit check. Taking $q = p$ in Lemma 1, we obtain

$$M_D\, \mathbb{E}|\widehat{z}_{xy} - z_{xy}|^2 \leq \frac{4}{s_e} \frac{M_D Q_{xy}}{q_x q_y} = \frac{4w_e^2}{s_e}, \qquad w_e = \sqrt{\frac{M_D Q_{xy}}{q_x q_y}}.$$

A comparison costs more if its check passes often relative to the squared signal $q_x q_y$, or if more mass inherits its error. Doubling its allocation s_e halves the upper bound.

Because $y \in D$ and x passes the check, $M_D \geq q_y$ and $Q_{xy} \geq q_x$. Thus $w_e \geq 1$. Round up to the next power of two (keeping an exact power unchanged), $r_e = 2^{\lceil \log_2 w_e \rceil}$ and write $W = \sum_e r_e$. For a nonempty list and a budget $B > 0$, allow positive real allocations with $\sum_e s_e = B$. Set $F = \sum_e r_e^2/s_e$. When $q = p$, $4F$ bounds the sum of the mass-weighted errors just derived. Several

errors can affect the same leaf. Lemma 5 will account for their accumulation using the maximum path length.

Cauchy–Schwarz states that $(\sum_e a_e b_e)^2 \le (\sum_e a_e^2)(\sum_e b_e^2)$, with equality for proportional nonzero vectors. Choose $a_e = \sqrt{s_e}$ and $b_e = r_e/\sqrt{s_e}$. Their product is the weight. Summing their squares gives the budget and the planning expression, so

$$W^2 \le \Big(\sum_e s_e\Big)\Big(\sum_e r_e^2/s_e\Big) = BF. \tag{1}$$

Equality requires s_e proportional to r_e, so $s_e = Br_e/W$ attains $F = W^2/B$. For a fixed target value of F, the required budget therefore grows as the square of the total weight. The next lemma chooses the references to keep that total small.

Lemma 3. Total weight of the comparisons. Given the tree of Lemma 2 and a positive distribution q on U, the references and intermediate labels can be chosen deterministically so that $q_{h_C} \ge M_C/(2|C|)$ and $q_g \ge M_D/(2|D|)$. All comparisons from C to its children have total rounded weight at most $12|C|$. Across the tree $W \le 12kL$, and any path uses at most $H = 2L$ comparisons.

The proof makes two choices. For the first comparison, we seek one parent reference with a small total cost to reach its children. Inside each child, we then choose an intermediate label frequent enough to make the second comparison affordable.

Proof. A singleton serves as its own reference. Fix an internal group C and abbreviate $m = |C|$, $M = M_C$, $t = t(C)$. We first choose one reference $h \in C$ for which the possible comparisons to labels $i \in C$ are affordable in aggregate. Any label in U passing the check on (h, i) satisfies the fixed tests for C, so we need only count labels in C. In these counts set $Q_{hh} = q_h$.

Form the coordinatewise majority string z, breaking ties as zero. Each $w \in C$ other than z differs from it at some bit. Mark one such bit for w. Charge its mass q_w to each label $i \in C$ on w's side of that bit, and call the total received by i its load L_i. At most t labels receive this charge, by the definition of t. If both endpoints were on the opposite side, their check would exclude w at its marked bit. Thus each passing marked label charges at least one endpoint. Consequently

$$\sum_{i \in C} L_i \le tM, \qquad Q_{hi} \le q_z + L_h + L_i,$$

where $q_z = 0$ if $z \notin C$.

If $q_z \ge M/m$, take $h = z$. This label receives no charge, and $\sum_i Q_{hi}/q_h \le m + tM/q_h \le m(t+1)$. Otherwise choose h minimizing

$$B_i = \frac{L_i + (t+1)M/m}{q_i}.$$

The average of B_i with weights q_i/M is at most $t + (t+1) = 2t+1$. Hence $B_h \le 2t+1$. Its constant numerator term guarantees $q_h \ge (t+1)M/[m(2t+1)] \ge M/(2m)$. Summing the passing-mass inequality and using $mq_z \le M$ also gives $\sum_i Q_{hi}/q_h \le [mL_h + (t+1)M]/q_h = mB_h$. Thus in either case $h = h_C$ meets the reference-mass bound and $\sum_i Q_{hi}/q_h \le m(2t+1)$.

In each child D of size d, choose g minimizing Q_{hi}/q_i among labels with $q_i \ge M_D/(2d)$. The excluded labels together have mass at most $M_D/2$. A weighted average over the eligible labels then gives $Q_{hg}/q_g \le 2\sum_{i \in D} Q_{hi}/M_D$. Put $A_D = \sum_{i \in D} Q_{hi}/q_h$. The first comparison thus has $w_{h,g} \le \sqrt{2A_D}$. For the second, both endpoints have mass at least $M_D/(2d)$, and $Q_{g,h_D} \le M_D$ since both lie in D. Hence $w_{g,h_D} \le 2d$. Fix all minimizing ties in label order.

There are $b \le 2m/t$ children by Lemma 2. Apply Cauchy–Schwarz to $a_D = 1$, $b_D = \sqrt{2A_D}$. Since $\sum_D A_D \le m(2t+1)$, the total raw weight out of C is at most

$$\sqrt{2b \sum_D A_D} + 2 \sum_D |D| \le m\sqrt{4(2t+1)/t} + 2m < 6m.$$

Rounding multiplies each weight by less than two, so the group costs at most $12m$. On one level the groups are disjoint. Adding their bounds gives $\sum_C 12|C| \le 12k$, since there are at most k labels in total. Summing over L internal levels yields $W \le 12kL$. At most two comparisons per descent give $H = 2L$. □

Now choose integer copy counts. To leave room for accumulation along paths, we make HF small compared with ε by setting

$$s_e = \lceil C_1 HW r_e/\varepsilon \rceil, \qquad H \sum_e r_e^2/s_e \le \frac{\varepsilon}{C_1} \sum_e \frac{r_e}{W} = \frac{\varepsilon}{C_1}. \tag{2}$$

This inequality applies when $W > 0$. An empty list uses zero phase copies. Fix $C_0 = C_1 = 2^{19}$ for the initial and phase batches, respectively. Lemmas 5 and 6 give their error guarantees. What remains is to build this plan from measurements when the probabilities are unknown.

4 Learning with unknown probabilities

We can now remove the assumption that the support and magnitudes are known. To plan each comparison, the learner needs the probabilities of its two endpoints and the probability of passing its agreeing-bit check. One batch of computational-basis measurements estimates both. We first ask how accurate those estimates must be to use the plan safely.

4.1 What one computational-basis batch reveals

Measure all qubits of N_0 independent copies in Z. Let U be the set of observed strings and q_x the fraction of copies that gave x, with $q_x = 0$ off U. The output magnitude for x will be $\sqrt{q_x}$. For a set A of strings, write $p(A) = \sum_{x \in A} p_x$ for its true probability. Its observed frequency $q(A) = \sum_{x \in A} q_x$ is the fraction of records landing in A. In particular, we can estimate a check's passing probability by applying its bit test to these same records.

Lemma 4. Accuracy of the initial Z measurements. For a distribution $p_x = |a_x|^2$ on at most k labels, put

$$\lambda = \left\lceil \log_2 \frac{8(k+k^2)}{\delta} \right\rceil, \qquad N_0 = \lceil C_0 k\lambda/\varepsilon \rceil, \qquad \tau = \lambda/N_0.$$

With probability at least $1 - \delta/4$, every true support singleton and every agreeing-bit set $A = \mathcal{B}_{xy}$ of two support labels satisfies $\left|\sqrt{q(A)} - \sqrt{p(A)}\right| \le \sqrt{\tau}$. Consequently $\sum_x (\sqrt{q_x} - \sqrt{p_x})^2 \le k\tau$.

The learner chooses the pairs after it sees U. To justify those choices, we prove the bound simultaneously for every pair it could choose.

Proof. Fix one such set A. Its count $N_0 q(A)$ is binomial with success probability $p = p(A)$. For $0 < p < 1$, the binomial Chernoff bound gives $\Pr(q(A) \ge u) \le e^{-N_0 D(u\|p)}$ for $u > p$, and the same

bound for $\Pr(q(A) \le u)$ when $u < p$. Here $D(u\|p)$ is Bernoulli relative entropy. To obtain a bound on square roots, write $v = (u, 1-u)$, $w = (p, 1-p)$. By concavity of the natural logarithm and $-\ln t \ge 1 - t$,

$$D(u\|p) = -2\sum_i v_i \ln\sqrt{w_i/v_i} \ge -2\ln\sum_i \sqrt{v_i w_i} \ge \sum_i (\sqrt{v_i} - \sqrt{w_i})^2 \ge (\sqrt{u} - \sqrt{p})^2.$$

Choose thresholds with $\sqrt{u} = \sqrt{p} \pm \sqrt{\tau}$, keeping only those in the range of a square root of a probability. Thus the two tails together have probability at most $2e^{-N_0\tau}$. Endpoints follow by continuity, and $p = 0, 1$ give exact frequencies. There are at most $k + k^2$ sets determined by the true support. Their total failure probability is at most $2(k+k^2)e^{-\lambda} \le \delta/4$. Squaring and summing the singleton bounds proves the magnitude claim. □

4.2 Which phases are worth measuring?

An *unseen* support label gets output amplitude zero. It contributes exactly p_x to the squared magnitude error in Lemma 4. An *observed* label keeps magnitude $\sqrt{q_x}$, but we can sometimes afford to give it an arbitrary phase. Two unit phases differ by at most two, so even a wholly wrong phase costs at most $4q_x$ in mass-weighted squared phase error.

Starting at the root, stop at the first group C on each path with average observed probability $M_C/|C| < 16\tau$. Omit the comparisons into and below this group and give its labels phase 1. The stopped groups are disjoint, so

$$\sum_{C \text{ stopped}} M_C \le 16\tau \sum_{C \text{ stopped}} |C| \le 16k\tau.$$

Their total squared phase cost is at most $64k\tau$. Stopping also protects the comparisons we keep. In a retained group the average is at least 16τ, and Lemma 3 picks endpoints of at least half the corresponding group average. Thus every measured endpoint has $q_x \ge 8\tau$. Below, W sums only the retained rounded weights. In the known-probability case take $q = p,\ \tau = 0$, so all groups are retained.

Lemma 5. Phase propagation and one-run accuracy. Fix frequencies q satisfying Lemma 4, or take the supplied probabilities. Use $s_e = \lceil C_1 HW r_e/\varepsilon \rceil$ fresh copies per retained comparison e, with no phase copies for an empty list. Give the root reference phase 1 and multiply the estimated ratios down each retained path. The normalized candidate ϕ satisfies

$$\mathbb{E}[1 - |\langle\psi, \phi\rangle|^2 \mid q] \le (130/C_0 + 384/C_1)\varepsilon.$$

The proof connects the actual measurements to the plan twice. First it compares each measured edge's true variance to its empirical estimate. Then it counts how much output mass uses that edge.

Proof. For each protected set with $q(A) > 0$ and $q(A) \ge 8\tau$, Lemma 4 puts $\sqrt{p(A)/q(A)}$ between $1 - 1/\sqrt{8}$ and $1 + 1/\sqrt{8}$. Hence $q(A)/4 \le p(A) \le 3q(A)$. Apply the lower bounds to the two endpoints. Their check contains both, so its empirical mass Q_{xy} also meets the threshold and its true mass P_{xy} is at most $3Q_{xy}$, including any unseen strings that pass. Consequently

$$\frac{P_{xy}}{p_x p_y} \le 48\,\frac{Q_{xy}}{q_x q_y}.$$

Choose the true global phase so its root reference has phase 1, and let $z_x, \widehat{z}_x$ be the true and estimated unit phases. For normalized vectors, $b = \langle \psi, \phi \rangle$ obeys $1 - |b|^2 \le 2(1 - \operatorname{Re} b) = \|\psi - \phi\|_2^2$. Inserting $\sqrt{q_x} z_x$ between the two amplitudes and using $|a+b|^2 \le 2|a|^2 + 2|b|^2$ bounds infidelity by

$$2 \sum_x (\sqrt{p_x} - \sqrt{q_x})^2 + 2 \sum_{x \in U} q_x |\widehat{z}_x - z_x|^2.$$

The magnitude term costs at most $2k\tau$. The stopped labels cost at most $2 \cdot 64 k\tau$ in the second term.

For a retained label x, replace the estimated ratios on its path $\pi(x)$ by their true values one at a time. All other factors have modulus one. By the triangle inequality and Cauchy–Schwarz, $(\sum a_i b_i)^2 \le (\sum a_i^2)(\sum b_i^2)$, with $a_e = 1,\ b_e = |\widehat{z}_e - z_e|$,

$$|\widehat{z}_x - z_x|^2 \le \left(\sum_{e \in \pi(x)} |\widehat{z}_e - z_e| \right)^2 \le H \sum_{e \in \pi(x)} |\widehat{z}_e - z_e|^2.$$

Multiply by q_x and sum over retained labels. A step $e = (x', y')$ into child D appears only on paths to labels in D, so it is charged at most M_D. Lemma 1 bounds its expected squared error by $4P_{x'y'}/(s_e p_{x'} p_{y'})$. Applying the factor 48 above and using $M_D Q_{x'y'}/(q_{x'} q_{y'}) \le r_e^2$ now gives

$$\mathbb{E}[1 - |\langle \psi, \phi \rangle|^2 \mid q] \le 130 k\tau + 384 H \sum_e r_e^2 / s_e \le 130 k\tau + 384 \varepsilon / C_1.$$

The last step is the allocation bound (2), including the empty sum. Finally $k\tau \le \varepsilon / C_0$. □

4.3 How to select an accurate candidate

With $C_0 = C_1 = 2^{19}$, Lemma 5 gives expected infidelity at most $\varepsilon/512$. Markov's inequality $\Pr(T > t) \le \mathbb{E}[T]/t$, applied to infidelity at $t = \varepsilon/64$, gives a candidate within this target with probability at least $7/8$ on a good initial batch. The learner can repeat the phase measurements. How can it choose among the candidates when the truth is unknown?

Lemma 6. Selecting by agreement between candidates. Fix a good initial batch. Make R independent phase runs, with R the smallest odd integer at least $4\lceil \log_2(2/\delta) \rceil$. For each candidate take the median of its squared trace distances to all R candidates, including itself. A candidate with smallest median attains infidelity at most ε with conditional probability at least $1 - \delta/2$, even if each squared distance has additive error at most $\varepsilon/1024$.

A strict majority of accurate candidates agree with each other. Any candidate that agrees with a strict majority must therefore agree with at least one accurate candidate.

Proof. For pure states the trace distance is $D(u, v) = \frac{1}{2}\|uu^* - vv^*\|_1 = \sqrt{1 - |\langle u, v \rangle|^2}$ since a nonzero difference of two pure-state projectors has nonzero eigenvalues $\pm\sqrt{1 - |\langle u, v \rangle|^2}$. The trace norm gives the triangle inequality. Call a run good if $D(\phi_i, \psi) \le \rho = \sqrt{\varepsilon}/8$. Conditional on the initial batch, the good-run indicators are independent and have mean at least $7/8$. Hoeffding's bound for independent indicators gives probability at most e^{-2Rt^2} of a downward deviation t of their average from its mean. Taking $t = 7/8 - 1/2$ bounds the chance of lacking a good majority by $e^{-2R(3/8)^2} \le \delta/2$.

When a strict majority is good, each good candidate has true median squared distance at most $4\rho^2$. An additive error $\eta = \varepsilon/1024$ in each distance squared changes a median by at most η. The

selected candidate therefore has true median at most $4\rho^2 + 2\eta$. Its ball of that squared radius contains a majority and meets the good set. The triangle inequality gives distance to ψ at most $\sqrt{4\rho^2 + 2\eta} + \rho < \sqrt{\varepsilon}$. The learner can compute all these distances from its records because the overlap of candidates i, j is $\sum_{x \in U} q_x \overline{\widehat{z}_{i,x}} \widehat{z}_{j,x}$. Adding the initial failure probability gives total failure at most $\delta/4 + \delta/2 < \delta$. □

5 The complete learner and its complexity

The learner first collects one batch in the computational basis. It can then write down the comparison pairs and copy counts for every subsequent run. Only the random choices of X or Y on individual copies and their measured outcomes vary from run to run.

Algorithm 1 Learning an unknown sparse pure state

Input. Independent copies of $|\psi\rangle$, the qubit count n, support bound k, infidelity target ε, and failure probability δ.

1. **Estimate the magnitudes.** Measure N_0 copies in Z on every qubit. Record the observed labels U and their frequencies q_x. Use N_0 and τ from Lemma 4.
2. **Choose the comparisons.** Construct the groups, references and comparisons of Lemmas 2 and 3 using U, q. Stop at the first group C on each path with $M_C < 16\tau|C|$. Omit comparisons entering or below a stopped group.
3. **Allocate copies.** Sum the retained weights to obtain W. Assign $s_e = \lceil C_1 HWr_e/\varepsilon \rceil$ fresh copies to comparison e in each of the R runs of Lemma 6.
4. **Estimate the phase ratios.** In each run and for each retained comparison, fix the bases on each copy and compute its score as in Lemma 1. Take the phase of the average of all s_e scores, choosing 1 for a zero mean.
5. **Construct a candidate.** Start the root reference at phase 1. Multiply the estimated phase ratios along each retained path. Give each stopped label phase 1. Multiply each unit phase by its corresponding magnitude $\sqrt{q_x}$.
6. **Select the estimate.** Compute each candidate's median squared trace distance to the R candidates. Choose a candidate with the smallest median, breaking ties by index.

Output. The chosen list of labels and complex amplitudes. All labels outside U have estimated amplitude zero.

Proof of Theorem 1. The output is normalized because each candidate has squared norm $\sum_{x \in U} q_x = 1$. Its list has at most k entries. Lemma 4 gives simultaneous accuracy of the initial data except with probability $\delta/4$. For every initial batch satisfying its bounds, Lemmas 5 and 6 give infidelity at most ε except with conditional probability $\delta/2$. Together these prove the stated success probability. It remains to count the copies and the work.

5.1 How many copies does the learner use?

The support bound k controls two different quantities. It bounds the number of comparisons, and it enters the precision required on those comparisons. To see the first, let the tree have $u = |U|$ leaves. Each internal group has at least two children, so there are at most $u - 1$ internal groups and $2u - 2$ parent–child links. At most two comparisons per link give at most $4(k - 1)$ comparison occurrences, even before stopping.

To see the precision cost, sum the allocations. A run assigns W shares in total. Equation (2) uses $C_1 HW/\varepsilon$ copies per share so that errors accumulated on paths fit the target. Multiplying the number of shares by the cost per share gives $C_1 HW^2/\varepsilon$. Rounding to whole copies adds at most one per comparison. With $W \le 12kL$ and $H = 2L$, the copies in one phase run satisfy

$$N_{\mathrm{ph}} = \sum_e s_e \le 4(k-1) + \frac{C_1 HW^2}{\varepsilon} \le 4(k-1) + \frac{288 C_1 k^2 L^3}{\varepsilon}.$$

Thus the second factor of k comes from squaring the bound on total weight. Two powers of L come from that square and one from the path length. The learner pays for the initial batch once and the phase measurements R times. Substituting $L = 1 + 2\lceil \log_2 k \rceil$ and $R = O(\log(2/\delta))$ gives the full copy bound

$$N = N_0 + RN_{\mathrm{ph}} = O\left(\frac{k}{\varepsilon} \log \frac{2k}{\delta} + \frac{k^2}{\varepsilon} \log^3(2k) \log \frac{2}{\delta}\right).$$

5.2 How much work does each stage require?

Plan the measurements. Sorting the N_0 observed strings and counting equal neighbors costs $O(nN_0 \log(2N_0))$ bit inspections. In a group of m labels, counting zeros and ones on every coordinate takes $O(nm)$ operations per split and $O(nm^2)$ over all its splits. To compute the loads in Lemma 3, visit each marked label and add its mass to the loads of labels on its marked side. After choosing h_C, there are only m candidates i for a first comparison. For each i, scan the m labels for agreement with (h_C, i) in $O(nm)$ operations. All observed labels passing this check are in C. Once the child references and intermediate labels have been selected, the second comparison for each child needs one scan of that child. The work per group is therefore $O(nm^2)$. At any level, $\sum_C |C|^2 \le (\sum_C |C|)^2 \le k^2$, giving $O(nk^2 L)$ operations for the tree and weights.

Measure copies and select the output. Generating bases, measuring the qubits, and computing the score take $O(n)$ elementary operations per phase copy. Multiplying phase ratios down the tree takes $O(k)$ arithmetic operations per run. A pair of candidates has at most k shared labels, so its overlap takes $O(k)$ operations. Compare all pairs and sort each row of distances to find its median. In total the operation count is

$$O\big(nN_0 \log(2N_0) + nk^2 L + nRN_{\mathrm{ph}} + kR^2 + R^2 \log(2R)\big).$$

Finite precision. Each observed mass is an integer count divided by N_0. The reference tests use integer cross-products. To round a weight, compare its squared rational value to powers of four. Each edge's score sum is a Gaussian integer (its real and imaginary parts are integers). Replace a zero sum by 1 and multiply the sums along the path to x to obtain B_x, using $B_x = 1$ for a stopped label. Its amplitude $\sqrt{q_x} B_x/|B_x|$ is an exact finite description. A path product uses $O(H \log(2N))$ bits.

For the classical selection step, compute each overlap summand to absolute error $\varepsilon/(8192k)$ by square-root bisection and rounding to a common binary grid. Summing the rounded terms incurs at most $\varepsilon/8192$ error in the overlap, whose true modulus is at most one. The resulting error in its squared modulus is less than $\varepsilon/1024$, as required in Lemma 6. With dyadic accuracy and confidence parameters (round each down within a factor two if necessary), a sufficient working bit length is $b = O(H \log(2N) + \log(2nk/\varepsilon) + \log(2/\delta))$. To see this explicitly, an overlap term has Gaussian-integer numerator and a denominator consisting of the count denominator and square roots of integer

squared norms. Comparing either component with a dyadic trial value requires only sign tests and squaring integers of $O(b)$ bits. Bisection for $O(b)$ steps with schoolbook arithmetic thus costs $O(b^3)$ bit operations per term. The same factor bounds the simpler arithmetic elsewhere. Multiplying the displayed operation count by this polylogarithmic overhead gives total time $\widetilde{O}(nk^2/\varepsilon)$. This completes the proof of Theorem 1.

The remaining question. How close is the quadratic copy bound to what is necessary? Even if the learner is given $k \geq 2$ possible labels, their span contains an arbitrary k-dimensional pure state. For pure states the trace distance is the square root of the infidelity. Taking rank one, dimension k, and trace-distance target $\sqrt{\varepsilon}$ in the lower bound of Scharnhorst, Spilecki, and Wright gives $\Omega(k/\varepsilon)$ copies at constant success probability for sufficiently small ε, even with unrestricted measurements [SSW25, Thm. 1.1]. Our quadratic upper bound comes from assigning a separate batch to each comparison and squaring their total weight. Whether single-qubit Pauli measurements can match both $\widetilde{O}(k/\varepsilon)$ copies and Sen's $\widetilde{O}(nk/\varepsilon)$ runtime for every support remains open [Sen26].

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